



Analyse Rev

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Génie Mécanique

1 Complexes

$$z^n - z = 0$$

$$r^n e^{in\varphi} = \omega_0 = s_0 e^{i\phi_0}$$

$$r = \sqrt[n]{s_0}, \varphi \in \left\{ \frac{2k\pi + \phi_0}{n} \mid (k \in [[0, n-1]]) \right\}$$

si

$$r \neq 0 \iff r^{n-1} e^{i(n+1)\varphi} = 1$$

$$r = 1, \varphi \in \left\{ \frac{2\pi k}{n+1} \mid (k \in [[0, n]]) \right\}$$

2 Suites

Limite

$$\frac{\sqrt[3]{n^3 + 8n^2 + 7} - n}{\frac{8n^2 + 7}{\sqrt[3]{n^3 + 8n^2 + 7}^2 + n^2 + n\sqrt[3]{n^3 + 8n^2 + 7}}}$$

3 Séries

$$\sum_{n=1}^{\infty} \left(1 + \frac{1}{n^2}\right)^{n^b} x^n$$

$$e^{n^{b-3}}$$

Pour $b = 2$ alors $e^1 R = e^{-1}$